A RAMCHARAN

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Professional Summary

Motivated Computer Science student with a strong willingness to learn and apply knowledge to real-world problems. Adaptable, responsible, and committed to continuous improvement, effective collaboration, and professional work practices. Brings a structured problem-solving approach, accountability, and a genuine interest in learning from challenges and feedback.

Technical Skills

Programming: Python, SQL, Java, C++, DataStructures
Machine Learning: Scikit-learn, Feature Engineering, Classification, Regression, Model Evaluation
Deep Learning: PyTorch, TensorFlow, Keras, CNN, BiLSTM, Transfer Learning, Data Augmentation
Generative AI: LangChain, RAG, LLMs, Prompt Engineering, OpenAI API, Google Gemini API, Hugging Face
Development: Flask, REST APIs, Git, GitHub, Docker, Linux/WSL, Postman
Data: NumPy, Pandas, Matplotlib

Projects

RAG Chatbot – Retrieval Augmented Generation **Jun 2026**
Python, Streamlit, LangChain, ChromaDB, Cohere, OpenAI, Google Generative AI, Hugging Face

- Designed a document-grounded chatbot to help users retrieve reliable answers from their own documents instead of relying only on general LLM knowledge.
- Processed PDF, TXT, DOCX, and CSV files using chunking, embeddings, ChromaDB, and semantic search for relevant information retrieval.
- Integrated contextual compression, Cohere reranking, and multiple LLM providers to improve response relevance and answer quality.

Saturn – NLP POS Tagging Model **Sep 2026**
Python, PyTorch, NLP, Word Embeddings, BiLSTM, DataLoader

- Stremlined a POS tagging model to automate grammatical analysis and improve contextual understanding of words in natural language.
- Trained on 12,543 sentences with 16,655 vocabulary tokens using 128-dimensional word embeddings with approximately 2.1M parameters.
- Implemented a BiLSTM with 128 hidden units per direction using PyTorch DataLoader for sequence-based training and contextual prediction.

Chest X-Ray Pneumonia Classification **Aug 2026**
Python, TensorFlow, Keras, MobileNetV2, Scikit-learn, NumPy, Pandas, Grad-CAM

- Created a classification system to assist in identifying pneumonia patterns in pediatric chest X-rays through automated medical image analysis.
- Preprocessed 224×224 X-ray images with augmentation and MobileNetV2 to classify images into NORMAL and PNEUMONIA categories.
- Achieved 78.04% test accuracy, 84.93% pneumonia F1-score, 94.69% ROC-AUC, and 98.97% pneumonia recall on 624 test images.

Internship

Machine Learning Intern | Pratinik Infotech India **Mar 2026 – May 2026**

- Gained practical experience solving real-world data problems while developing a responsible, structured, and analytical approach to machine learning tasks.
- Engineered features across 10+ datasets, reducing preprocessing time by 25% and improving efficiency in ML workflows.
- Developed a Random Forest predictive maintenance model achieving 82% accuracy and fraud detection classifiers achieving 75% precision across 100+ records.

Certifications

- Azure AI Engineer Associate (AI-102) — Microsoft — Mar 2026
- Azure AI Fundamentals (AI-900) — Microsoft — Dec 2025
- Azure Data Fundamentals (DP-900) — Microsoft — Mar 2025
- Azure Fundamentals (AZ-900) — Microsoft — Dec 2024

Education

Lovely Professional University **Aug 2024 – Jun 2028**
B.Tech in CSE (Generative AI), 9.22 CGPA *Jalandhar, Punjab*

Sai Sri Chaitanya Junior College **2022 – 2024**
Mathematics, Physics & Chemistry, 95.3% *Andhra Pradesh, India*

Achievements

- Led teams in 3+ hackathons, qualified for 2 rounds, and participated in national-level hackathons at IIT Jodhpur and IIT Ropar.
- Strengthened teamwork, problem-solving, communication, and rapid decision-making while developing solutions under time and resource constraints.